

5. Simulate bouncing ball in Unity. The bouncing should stop gradually after the time specified by the user.

I'll guide you step-by-step so your ball:

- Bounces realistically 🏐
- Gradually stops ☐
- Stops after **user-defined time** ⌚

🎯 FINAL RESULT

- Ball drops → bounces → loses energy → stops
- User controls **how long it keeps bouncing**

☐ STEP 1: Create Scene

✓ Ground

1. Right-click → **3D Object** → **Plane**
2. Rename → `Ground`

✓ Ball

1. Right-click → **3D Object** → **Sphere**
2. Rename → `Ball`
3. Position:

X: 0
Y: 5 ← IMPORTANT (height for bouncing)
Z: 0

⚙️ STEP 2: Add Physics

Select **Ball**:

Add:

- ✓ Rigidbody

Rigidbody Settings:

Mass: 1
Drag: 0
Angular Drag: 0.05
Use Gravity: ✓

□ STEP 3: Create Bouncy Material

1. Right-click → **Create** → **Physic Material**
 2. Name → BouncyMat
-

🔧 Set Values:

Bounciness: 0.8
Bounce Combine: Multiply

Apply it:

- Drag BouncyMat → onto **Ball**
- Also drag it → onto **Ground**

☞ Now ball will bounce

📄 STEP 4: Create Script

Create script → BounceController

```
using UnityEngine;

public class BounceController : MonoBehaviour
{
    public Rigidbody rb;

    public float stopTime = 5f; // ☞ USER CONTROL
    public float damping = 0.98f;

    private float timer = 0f;
```

```
void Update()
{
    timer += Time.deltaTime;

    // Gradually reduce bounce effect
    if (timer < stopTime)
    {
        rb.linearVelocity *= damping;
    }
    else
    {
        StopBall();
    }
}

void StopBall()
{
    rb.linearVelocity = Vector3.zero;
    rb.angularVelocity = Vector3.zero;

    // Freeze movement completely
    rb.isKinematic = true;
}
}
```

STEP 5: Attach Script

1. Select **Ball**
 2. Drag `BounceController` script onto it
-

STEP 6: Assign Rigidbody

In Inspector:

- Drag **Ball** → into `Rb` field
-

STEP 7: Set Stop Time

In Inspector:

Stop Time = 5

 You can change:

- 3 → stops quickly
 - 10 → bounces longer
-

► STEP 8: Run

1. Click **Play**

☞ You'll see:

- Ball falls
 - Bounces
 - Gradually slows
 - Stops after given time
-

⚠ Important Concept (for viva)

☞ Why it stops gradually?

Because:

Velocity is reduced every frame using damping

Correct Way to Apply **BouncyMat**

◆ Step 1: Select the Ball

1. Click **Ball** in Hierarchy
2. In Inspector, find:

Sphere Collider

◆ Step 2: Apply Material to Collider

Inside **Sphere Collider**, you'll see:

Material [None]

☞ **Drag BouncyMat into this "Material" field**

◆ Step 3: Do the same for Ground

1. Select **Ground**
2. Find:

`Mesh Collider`

3. Drag `BouncyMat` into:

`Material field`

! Why your drag didn't work

You were probably trying to:

- Drag onto object ✗
- Drag onto Mesh Renderer ✗

☞ **Physic Material ONLY** works with:

`Collider component` ✓

🔍 If you don't see "Material" field

For Ball:

- Must have **Sphere Collider** ✓

For Ground:

- Must have **Mesh Collider** ✓
(Plane already has it by default)
-

🔑 Quick Visual Checklist

Ball Inspector should show:

Sphere Collider
Material → BouncyMat ✓

Ground Inspector:

Mesh Collider
Material → BouncyMat ✓

1. Increase Bounciness (MOST IMPORTANT)

Select your **BouncyMat**:

Set:

Bounciness = 0.9 → try 1.0
Bounce Combine = Maximum

☞ This directly controls how high the ball rebounds

✓ 2. Apply Material Correctly (Double Check)

Make sure BOTH have it:

Ball:

- Sphere Collider → Material = BouncyMat

Ground:

- Mesh Collider → Material = BouncyMat
-

✓ 3. Reduce Energy Loss

Select **Ball** → **Rigidbody**:

Set:

Drag = 0
Angular Drag = 0.05 (or 0)

☞ High drag = less bounce

✓ 4. Increase Drop Height

Select Ball → Position:

`Y = 5` → try 8 or 10

☞ More height = more bounce energy

✓ 5. Check Gravity (important)

In Rigidbody:

`Use Gravity = ✓`